MIL-P-36358B
6 October 1976
SUPERSEDING
MIL-P-36358A(DSA-DM)
28 October 1970

MILITARY SPECIFICATION

PAD, HEAT TREATMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a waterproof, electric heating pad suitable for use in military medical treatment facilities.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq., Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Phila. PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6530

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

UNDERWRITERS' LABORATORIES, INC. (UL)

UL 130 Standard for Electric Heating Pads

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 1285 Walt Whitman Road, Melville, L.I, NY 11749; 207 East Ohio Street, Chicago, IL 60611, 1655 Scott Boulevard, Santa Clara, CA 95050 and 333 Pfingsten Road, Northbrook, IL 60062.)

NATIONAL BUREAU OF STANDARDS

NBS Monograph 90 Calibration of Liquid-in-glass Thermometers

(Application for copies should be addressed to Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

3. REQUIREMENTS

* 3.1 First article. When specified in the procurement document, the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.1.2.1 and 6.2).

3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of Underwriters' Laboratories, Inc. (UL) Standards of Safety UL 130. Proof shall consist of one of the following:

A. UL label affixed to the product, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

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B. Laboratory report from a nationally recognized independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The Government reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.3 Material. Material used in the fabrication of the pads shall be as specified herein, of the highest quality throughout, and suitable in all respects for the purpose intended.

3.4 Construction. The heating pad shall be a conventional type consisting basically of a thermoplastic material envelope such as a vinyl cover, which is used to enclose a resistance wire heating element assembly. The heating element shall be mounted on a layer of burlap reinforced cotton insulating material and shall be positioned in such a manner as to distribute the resulting heat evenly over both sides of the pad. Wetproof construction shall be provided by means of the envelope which shall have dielectrically or heat sealed seams to exclude moisture. The waterproof envelope shall be free of perforations, cuts, tears and ruptures. Further, it shall not exhibit tackiness, blisters or cracks. Thermostats, of a type designed for minimum radio interference, shall be provided to limit the maximum temperatures reached by the heating element. A single thermostat connected in series with the heating element is not acceptable. The pads when tested as specified in 4.4.1 or 4.4.1.1 shall maintain the temperature within 48° to 55°C. at any exterior point within a rectangular area inscribed at least 2 inches from each side of the pad. The circuitry shall be such that the thermostat shall shut off the current before the temperature of the pad exceeds the required setting. The heating pad shall be provided with not less than 6 feet of a suitable SJT, or heavier flexible two conductor power cord and a molded two prong plug. Strain relief shall be provided on all cords to prevent any mechanical stress on the cord from being transmitted to any wiring connections. The pad shall operate satisfactorily on 115 volts, 60 Hz, AC and shall be rated at 55-60 watts. Overall dimensions of the pad shall be 12 by 15 inches. Each heating pad shall be furnished with two washable easily removable terrycloth covers with four 16 inch tie straps on each cover.

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* 3.5 Identification marking. Each unit shall have one or more labels securely attached to the envelope marked permanently and legibly with the following information:

Name or registered trademark of the manufacturer
Model number
Electrical rating
Contract number or lot number

3.6 Instructions. The pad shall have the following warning, or equivalent, marked on the face of the pad:

WARNING
BURNS WILL RESULT FROM IMPROPER USE.
NEVER USE PAD WITHOUT COVER IN PLACE.

CAREFULLY EXAMINE BEFORE EACH USE.
FOLLOW INSTRUCTIONS SUPPLIED WITH THE PAD.
DO NOT USE IN AN OXYGEN ATMOSPHERE.
DO NOT USE PINS.

Each pad shall have the following printed instructions, or equivalent, packed with the pad:

INSTRUCTIONS IMPORTANT. DO NOT DESTROY.

OBSERVE THESE INSTRUCTIONS CAREFULLY TO REDUCE THE POSSIBILITY OF FIRE OR ACCIDENT:

DO NOT USE IN OXYGEN ATMOSPHERE.
TEMPERATURES SUFFICIENTLY HIGH TO CAUSE BURNS MAY OCCUR REGARDLESS OF THE CONTROL SETTING. DO NOT USE ON AN INFANT OR INVALID OR ON A SLEEPING OR OTHERWISE UNCONSCIOUS PERSON UNLESS CAREFULLY ATTENDED. DO NOT USE ON INSENSITIVE SKIN OR A PERSON WITH POOR CIRCULATION. CHECK THE SKIN UNDER THE PAD FREQUENTLY TO AVOID BURNING AND BLISTERING.

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DO NOT REST ANY PART OF THE BODY ON PAD.
BURNS CAN RESULT REGARDLESS OF CONTROL SETTING.

CAREFULLY EXAMINE THE PERMANENT INNER COVERING BEFORE EACH USE. DISCARD THE PAD IF THE COVERING SHOWS ANY SIGNS OF DETERIORATION (SUCH AS CHECKING, BLISTERING OR CRACKING).

NEVER USE PINS OR OTHER METALLIC MEANS TO FASTEN THIS PAD IN PLACE.

DO NOT CRUSH. AVOID SHARP FOLDS.

NEVER PULL THIS PAD BY THE SUPPLY CORD AND DO NOT USE THE CORD AS A HANDLE.

UNPLUG WHEN NOT IN USE.

3.7 Workmanship. Workmanship shall be first class throughout. Pads shall be free from defects such as dirt, stains or discoloration which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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* 4.1.2.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection.
b. Quality conformance inspection.

* 4.1.2.1.1 First article inspection. First article inspection shall be performed on the pad when a first article sample is required (see 3.1). This inspection shall include examination of 4.3 and tests of 4.4.

* 4.1.2.1.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.3 and tests of 4.4.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I. The unit of product for sampling purposes shall be one pad.

TABLE I. Sampling for examination.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	1.0
Total defects (Major and Minor combined)	II	4.0
For dimensional examination	II	2.5

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4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table II, for an Inspection level of S-1 and AQL of 0. The unit of product for sampling purposes shall be one pad.

TABLE II. Sampling for tests.

Component (Unit of product)	Characteristic	Requirement	Test procedure
Heating circuit	Temperature control	3.4	4.4.1 or 4.4.1.1

4.2.3 Internal construction. Two heating pads from each inspection lot shall have the plastic envelope and cotton insulation removed from the heating element. Insulating tape shall be removed from electrical connections. The construction and workmanship shall be examined. Presence of defects which may impair serviceability, or show evidence of poor workmanship shall be cause for rejection.

4.3 Examination. The pad shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the classification of defects listed in table III. Examination shall not be restricted to the classified possible defects listed below.

TABLE III. Classification of defects.

Categories	Defects
Major	
101	Pad not complete with all components.
102	Pad not equipped with warning instructions.
103	Warning instructions not legible, nor complete.
104	Waterproof envelope not free of perforations, cuts, tears, ruptures.
105	Waterproof envelope not free of tackiness, blisters, cracks.
106	Supply cord or plug not as specified.
107	No strain relief at both ends of power cord.
108	Insulation not intact on power cord.
Minor	
201	Instructions for use not packaged with pad, not legible, or not complete.
202	Identification marking not on pad, not legible, or not complete.
203	Unit not free of dirt, stains, discolorations.

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4.3.2 Dimensional examination. The pad shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Temperature control test. Starting with a pad at room temperature, it shall be connected to a supply circuit of maximum voltage. The pad shall be laid flat between two felt mats, each one inch thick, and with an area sufficient to cover the pad completely with a margin of not less than two inches all the way around. The pad shall be operated completely covered, for a period of four hours, or longer, if necessary, to reach thermal equilibrium. During this test, temperatures are to be observed on all mercury glass thermometers which have been calibrated in accordance with NBS Monograph 90 at not less than six points on the exterior of the pad. Use of thermocouples in lieu of thermometers is permitted; however, thermocouples shall be calibrated to an NBS thermometer prior to use. Twenty minutes after the first opening of a thermostat and subsequently, the temperatures of any thermometer must be within the range specified in 3.4. After disconnecting the pad from the supply circuit and removing the felt mats, the pad shall be allowed to return to thermal equilibrium at room temperature as indicated by the six thermometers.

* 4.4.1.1 Alternate temperature control test. Starting with the pad at room temperature, maximum power shall be applied to the pad. The pad shall be placed between two layers of thermal insulating material such as blankets or felt padding which completely covers the pad. A mercury thermometer calibrated in accordance with NBS Monograph 90 shall be taped to a 3 inch square of light sheet metal or brass shim. Use of thermocouples in lieu of thermometers is permitted; however, thermocouples shall be calibrated to an NBS thermometer prior to use. The metal shall be taped to the center of the pad (with the metal between the thermometer and the pad). Allow the system to stabilize for a minimum of ten minutes prior to taking readings. No reading shall be outside the temperature range as indicated in 3.4. After disconnecting the pad from the supply circuit and removing the thermal insulating material (blankets or felt padding), the pad shall be allowed to return to thermal equilibrium at room temperature as indicated by the thermometer.

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4.5 Packaging inspection. The inspection of the unit, intermediate package and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One pad, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each pad, with the cord neatly arranged, shall be inserted into a sealed polyethylene bag. Each pad shall be further packaged in a box conforming to PPP-B-676. Closure of the box shall be as specified in the appendix of the box specification.

5.1.3 Level C. Each pad shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing.

5.2.1 Level A. Twelve unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twelve unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged pads shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. Each unit package and exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The electric heating pad is intended for the application of heat to small areas of the body.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents.

a. Title, number and date of this specification.
b. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
c. When a first article is required (see 3.1, 4.1.2.1 and 6.3).

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of the Armed Services Procurement Regulations. The contract or purchase order specifies the quantity, procedure for delivery and other specific instructions for inspection and approval of the first article.

6.4 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock NUmber	Item Identification
6530-01-015-5168	Pad, Heat Treatment, 115 volt, 60 Hz, AC

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
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Preparing activity:
DSA-DM

Project No. 6530-1400

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DOCUMENT IDENTIFIER AND TITLE

MIL-P-36358B PAD, HEAT TREATMENT

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

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